

Discrete Probability Distributions

Discrete distributions model outcomes that are **countable** (integers).

1. Bernoulli Distribution $Bern(p)$

- **Definition:** Represents a single trial with only two possible outcomes: **Success** ($X = 1$) or **Failure** ($X = 0$).
- **Parameter:** p , the probability of success.
- **Example:** A single coin flip, or checking a single item on an assembly line for a defect.
- **Probability Mass Function (PMF):**

$$P(X = k) = p^k(1 - p)^{1-k} \quad \text{for } k \in \{0, 1\}$$

- **Key Insight:** It is the foundation for almost all other discrete distributions.
- **Key Statistics:**
 - $E[X] = p$
 - $\text{Var}(X) = p(1 - p)$

2. Binomial Distribution $Bin(n, p)$

- **Definition:** Represents the number of successes in a fixed number, n , of **independent and identical Bernoulli trials**, where p is the probability of success in each trial.
- **Parameters:** n (number of trials) and p (probability of success).
- **Example:** Flipping a coin 10 times and counting the number of heads; counting the number of defective items in a sample of 100.
- **Probability Mass Function (PMF):**

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \quad \text{for } k \in \{0, 1, \dots, n\}$$

- **Key Insight:** The $\binom{n}{k}$ term is the number of ways to arrange k successes and $n - k$ failures.
- **Key Statistics:**
 - $E[X] = np$
 - $\text{Var}(X) = np(1 - p)$

3. Poisson Distribution $Pois(\lambda)$

- **Definition:** Models the **number of events** occurring within a fixed interval of time or space, given a known average rate of occurrence, λ (lambda).
- **Parameter:** λ , which is the average rate of events.
- **Example:** The number of customers arriving at a store in an hour, the number of emails received in a day, or the number of typos on a page.
- **Probability Mass Function (PMF):**

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!} \quad \text{for } k \in \{0, 1, 2, \dots\}$$

- **Key Insight:** As you noted, it's the limit of the Binomial distribution when n is very large and p is very small, such that $np = \lambda$.
- **Key Statistics:**
 - $E[X] = \lambda$
 - $\text{Var}(X) = \lambda$ (This equality is a hallmark of the Poisson distribution.)

4. Geometric Distribution $Geom(p)$

- **Definition:** Models the number of **independent Bernoulli trials required to get the first success**.
- **Parameter:** p , the probability of success on a single trial.
- **Example:** The number of times a basketball player shoots until they make their first basket.
- **Probability Mass Function (PMF):**

$$P(X = k) = (1 - p)^{k-1} p \quad \text{for } k \in \{1, 2, 3, \dots\}$$

- **Key Property: Memoryless Property** in the discrete domain. The probability of needing to wait t more trials for a success, given that you have already waited s trials with no success, is the same as the probability of needing to wait t trials from the start.

- **Key Statistics:**
 - $E[X] = 1/p$
 - $\text{Var}(X) = (1-p)/p^2$

5. Hypergeometric Distribution $HGeom(N, K, n)$

- **Definition:** Models the number of successes in a sample of size n , drawn **without replacement** from a finite population of size N that contains K successes.
- **Parameters:** N (population size), K (number of successes in the population), n (sample size).
- **Example:** Drawing winning lottery tickets from a finite pool of tickets where some are winners, without putting the tickets back. Quality control inspection where items are removed from the batch.
- **Probability Mass Function (PMF):**

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

- **Key Insight:** It is the correct model when the trials are **not independent** (because sampling without replacement changes the success probability for subsequent draws).
- **Key Statistics:**
 - $E[X] = n \cdot \frac{K}{N}$
 - $\text{Var}(X) = n \frac{K}{N} \frac{N-K}{N} \frac{N-n}{N-1}$ (Note the $\frac{N-n}{N-1}$ term, called the **finite population correction factor**, which accounts for sampling without replacement.)

Continuous Probability Distributions

Continuous distributions model outcomes that can take **any value** within a specified range (real numbers).

[Image of three continuous probability density functions]

1. Uniform Distribution $U(a, b)$

- **Definition:** All values within a given interval $[a, b]$ are **equally likely**.
- **Parameters:** a (minimum value) and b (maximum value).
- **Example:** A random number generator that selects any real number between 0 and 1.
- **Probability Density Function (PDF):**

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

- **Cumulative Distribution Function (CDF):** The CDF gives the probability $P(X \leq x)$.

$$F(x) = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } a \leq x \leq b \\ 1 & \text{for } x > b \end{cases}$$

- **Key Statistics:**
 - $E[X] = \frac{a+b}{2}$
 - $\text{Var}(X) = \frac{(b-a)^2}{12}$

2. Exponential Distribution $Exp(\lambda)$

- **Definition:** Models the **waiting time** until the next event in a Poisson process. It is used for modeling lifetime/failure time.
- **Parameter:** λ , the **rate** of the events (the inverse of the mean waiting time).
- **Example:** The lifespan of a non-aging machine component, or the time until the next phone call is received at a call center.
- **Probability Density Function (PDF):**

$$f(x) = \lambda e^{-\lambda x} \quad \text{for } x \geq 0$$

- **Key Property: Memorylessness.** It is the *only* continuous distribution with this property.
- **Key Statistics:**

- $E[X] = 1/\lambda$
- $\text{Var}(X) = 1/\lambda^2$

3. Normal (Gaussian) Distribution $N(\mu, \sigma^2)$

- **Definition:** The **most common and important** distribution. It describes data that clusters around a central mean value, forming a symmetric, bell-shaped curve.
- **Parameters:** μ (mean/center) and σ^2 (variance/spread).
- **Example:** Human height, blood pressure, measurement errors in experiments, financial stock returns.
- **Probability Density Function (PDF):**

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

- **Key Property: The Central Limit Theorem (CLT):** This is *why* the Normal distribution is ubiquitous. The CLT states that the **distribution of the sample means** of many independent random variables will approach a Normal distribution, regardless of the shape of the original population distribution.
- **Standard Normal Distribution:** $N(0, 1)$ with $\mu = 0$ and $\sigma^2 = 1$. Any Normal variable X can be converted to the Standard Normal variable Z using the **Z-score** formula: $Z = \frac{X-\mu}{\sigma}$.
- **Empirical Rule (68-95-99.7 Rule):** Approximately 68% of the data falls within $\pm 1\sigma$ of the mean, 95% within $\pm 2\sigma$, and 99.7% within $\pm 3\sigma$.

Special/Derived Continuous Distributions

4. Chi-Squared $\chi^2(k)$

- **Definition:** The distribution of the **sum of the squares of k independent Standard Normal random variables**.
- **Parameter:** k , the **degrees of freedom**.
- **Usage:** It is the foundation for χ^2 tests (for independence and goodness-of-fit) and for statistical inference about the **population variance**.

5. Student's t -Distribution $t(k)$

- **Definition:** Arises when estimating the **mean of a normally-distributed population when the sample size is small** and the population standard deviation (σ) is **unknown**.
- **Parameter:** k , the **degrees of freedom** (usually $n - 1$, where n is the sample size).
- **Key Insight:** It is **bell-shaped and symmetric** like the Normal distribution, but has **heavier tails** (more probability in the extremes). As $k \rightarrow \infty$, the t -distribution converges to the Standard Normal distribution.
- **Usage:** Used for t -tests and constructing confidence intervals for the population mean.

6. F-Distribution $F(k_1, k_2)$

- **Definition:** The distribution of the **ratio of two independent Chi-Squared variables**, each divided by its degrees of freedom.
- **Parameters:** k_1 and k_2 , the degrees of freedom for the numerator and denominator, respectively.
- **Usage:** Primarily used in **Analysis of Variance (ANOVA)** to compare the means of three or more populations, and for testing the **equality of two population variances**.

Would you like a more detailed breakdown of the relationships *between* these distributions (e.g., how the Exponential and Poisson are linked, or the Binomial and Normal)?